



Aalto University
School of Chemical
Engineering

CS-C3240 Machine Learning

Modelling crop yields based on early season weather, soil, and growth data

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Aalto University

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1. Problem formulation

A model can be trained to predict the crop yield of a specific crop based on early-season (Apr-Jul) conditions. While a regression model is best for this task, the question whether to regularise the model arises.

A dataset on CONUS (contiguous United States) agricultural details was utilised [1], [2]. This dataset includes measurements on corn, wheat, and soy yields, as well as corresponding values for precipitation (monthly), soil moisture (daily), maximum temperature (monthly), VOD (vegetative optical depth, daily), and EVI (enhanced vegetation index, daily). All measurements are continuous in value. Soybean was chosen as the crop of interest for this model as it is a widely grown crop for both human consumption and animal feed.

From this dataset, datapoint labels were individual yields for a specific county and a specific year. These were used for a supervised learning application. The years measured were 2015-2018, and there was a datapoint for each of the counties measured for each year. Each datapoint used the soybean crop yield in t/ha as a label. 12 features were chosen for the model, with dates ranging from April to end of July:

1. Maximum temperature for each month [$^{\circ}\text{C}$]
2. Total precipitation for each month [mm]
3. Arithmetic average of soil moisture measurements [$\text{m}^3 \cdot \text{m}^{-3}$]
4. The 95th percentile and 5th percentile of soil moisture measurements [$\text{m}^3 \cdot \text{m}^{-3}$]
5. Standard deviation of VOD

2. Methods

The dataset contains 2060 datapoints, split into 515 datapoints per year, one per each county. Given the even spread of 4 datapoints per county, all datapoints were treated equally. First the data was cropped to include values only for the time period from start of April to end of July. Maximum temperature measurements and precipitation measurements were combined into one csv file with the label vector. Soil moisture and VOD were processed as separate csv files due to the large number of measurements.

From the features, a dataframe containing values for all precipitation and maximum temperature values for the months April-July, different statistical values thereof, and several different statistical values of the SM and VOD datasets was constructed. A pearson cross-correlation matrix was created for this dataframe, shown in Figure 1.

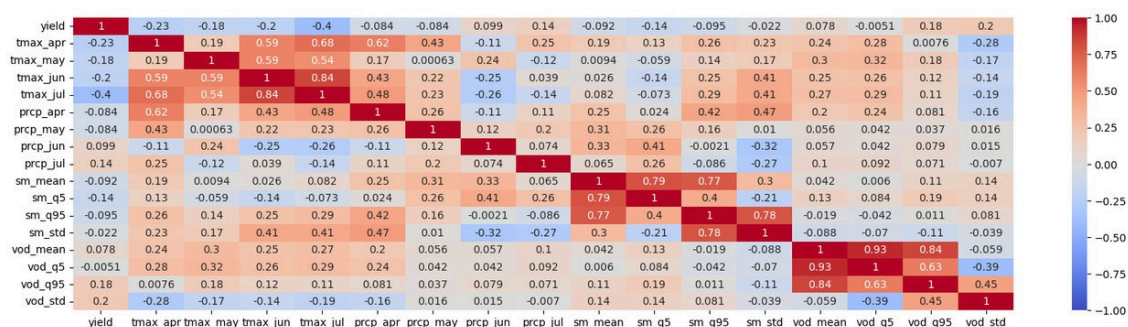


Figure 1, Pearson cross-correlation matrix of measurement values for feature selection.

The matrix in Figure 1 shows low correlations for many of the potential features with the label ‘yield’.

VOD values were selected to be represented by the standard deviation over the four months. This was done due to their high correlation with each other as well as low correlation with the label. In addition, mean values for VOD could be misleading since existing permanent vegetation in the measurement area could skew the values. The change in VOD was considered to represent the change in crop biomass most accurately. Maximum temperature values were chosen to be kept as-is despite their high co-linearity. The response of soybean growth to temperature fluctuations is highly different in different stages of the crop’s growth cycle [3]. All temperature values also show a rather high correlation with the label compared to other features. Precipitation values were all kept. They have a low co-linearity and acceptably low values for cross-correlation with other features. Soil moisture was chosen to be represented by three values: the arithmetic mean, the 5th percentile, and the 95th percentile. The mean represents overall differences in average soil moisture. The 5th and 95th percentiles take into account exceptionally dry and humid conditions, respectively, while being slightly less susceptible to extreme outliers. The selected features and their cross-correlations are shown in Figure 2.

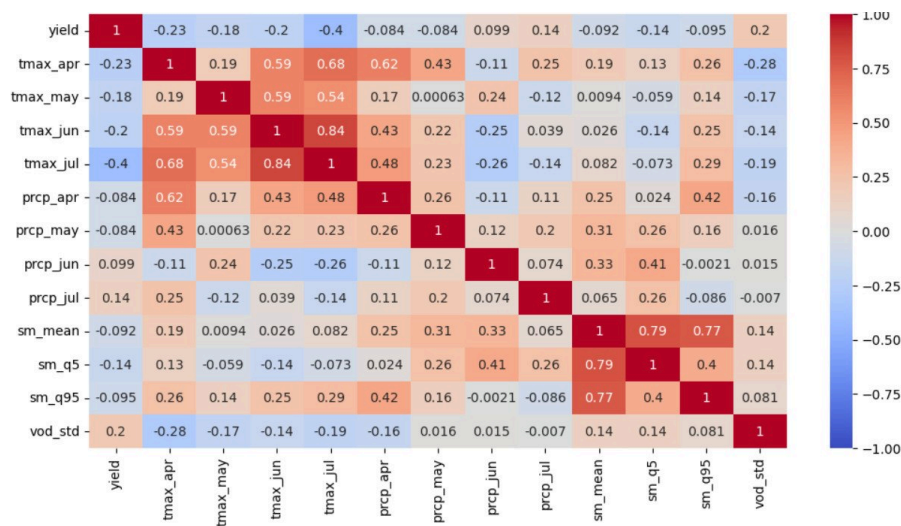


Figure 2, Pearson cross-correlation matrix of chosen features.

As the label is continuous in value, a regression model was thought to be most suitable for this problem. The co-linearity of some features prompted the the consideration of a model with regularisation. The featured for maximum temperature values and for soil moisture values have strong cross-correlation, but they all still affect crop yields in different ways. A Ridge regression was chosen to account for this problem specifically.

Linear regression is used to predict continuous values based on the chosen features. It outputs a continuous variable calculated on the features, with each feature having a specific weight coefficient. A common method to determine the coefficients is the squared error. The issue with an ordinary squared error loss function is the blow-up of coefficients: especially in cases, where some features exhibit co-linearity, they may be assigned extremely large coefficients correcting for each other, which leads to overfitting and high sensitivity.

Ridge regression adds a penalty term to the squared error loss function based on the square of the coefficients, unlike L1 style regularisation methods such as LASSO. This ensures that no features are regularised to zero coefficients and thus maintains all feature data, while also effectively controlling the scale of the coefficients. The result is an L2 regularised mean squared error loss function.

All features were normalised using StandardScaler, because some features are on vastly different scales. For the Ridge regression's regularisation to treat all features equally, they must have 0 mean and unit variance.

Data was split into four sections by year. The latest year was chosen as the testing set. The remaining three years of data from 2015 to 2017 will be used as data in a k-fold cross validation method with $k = 3$, each fold being one year, to find the optimal Ridge parameter α . Values for this hyperparameter were chosen tested over a logarithmic range of 50 values from 0.001 to 100 000.

After the optimal α is determined, the whole training set of data from 2015 to 2017 will be used to train the Ridge regression model.

3. Use of AI

Several large language models by Anthropic (Claude) were utilised in the making of this project for assistance and automation in the following tasks:

1. Comparing different data from the original datasets to determine which would show valuable information as features, including but not limited to explanations of concepts such as VOD (vegetation optical depth) and EVI (enhanced vegetative index).
2. Finding methods for different tasks to be completed on the data set from the PANDAS and NUMPY documentations.
3. Error checking and code validation.

4. Bibliography

- [1] L. Martinez-Ferrer, M. Piles, and G. Camps-Valls, "Crop Yield Estimation and Interpretability With Gaussian Processes," *IEEE Geoscience and Remote Sensing Letters*, vol. 18, no. 12, pp. 2043–2047, Dec. 2021, doi: 10.1109/LGRS.2020.3016140.
- [2] A. Mateo-Sanchis, J. E. Adsua, M. Piles, J. Munoz-Mari, A. Perez-Suay, and G. Camps-Valls, "Interpretable Long Short-Term Memory Networks for Crop Yield Estimation," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3244064.
- [3] J. L. Hatfield and J. H. Prueger, "Temperature Extremes: Effect on Plant Growth and Development," *Weather and Climate Extremes*, vol. 10, pp. 4–10, Dec. 2015, doi: 10.1016/j.wace.2015.08.001.

5. Appendices

The code for this model can be found on Github in the file 'regression_3.ipynb' by following this [link](#).